

Hubungi

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Keahlian Teratas

Earth Engine APPS
Python (Programming Language)
Data Visualization

Languages

Bahasa Jawa (Elementary)
Bahasa Arab (Limited Working)
Bahasa Inggris (Elementary)
Bahasa Indonesia (Native or Bilingual)
Sundanese (Native or Bilingual)

Certifications

Intermediate use of Google Workspace for Education
How to manage and publish your research data
Processing SAR Data in ArcGIS Notebooks
Microsoft Educator Trainer Academy
Planetary Scale Earth Observation with Google Earth Engine

Defani Arman Alfitriansyah

Final Year Forestry Student | Remote Sensing Analyst | Ecological Modeling | Forest Carbon | Python & R Programming Enthusiast
Kuningan, Jawa Barat, Indonesia

Ringkasan

I am a final-year Forestry student at Universitas Kuningan focusing on remote sensing for forestry, particularly carbon modeling and ecological modeling. My research involves carbon stock estimation using Sentinel-2 indices and developing a Google Earth Engine (GEE) application for mangrove above-ground biomass monitoring in Subang Regency.

I completed an internship at the Forum Pohon Langka Indonesia (FPLI) as a rare tree surveyor and contributed to nursery management for conservation and restoration programs. I also worked on a remote sensing project identifying priority areas for ecological restoration, served as an ecology trip guide, and delivered training on mapping workflows for rare tree surveys. Additionally, I was invited as a speaker in a summer camp to educate children about forest functions and environmental conservation.

Previously, I joined the MBKM internship with the Ministry of Environment and Forestry (KLHK) as a Social Forestry Facilitator in North Sumatra. I assisted community-based forest management and developed a spatial database for HKM Muara Indah Jaya Mandiri to support planning and monitoring. I am also active in the Conservation Study Group (KSK), contributing to research, field activities, and scientific discussions.

My experience includes mangrove carbon surveys, GIS and Spatial Modeling Laboratory Assistant, and teaching private GIS and remote sensing courses. I also create educational content about GIS, remote sensing, and ecological data analysis using GEE, geemap, Python, and R.

I am motivated to collaborate on research and contribute to sustainable forest management through geospatial technology and ecological studies.

Research Interests: Remote sensing for forestry, forest carbon modeling, biomass estimation, mangrove ecosystems, ecological modeling, restoration prioritization, biodiversity conservation, social forestry, environmental education, and geospatial applications for sustainability.

Pengalaman

Badan Riset dan Inovasi Nasional RI (BRIN RI)

Remote Sensing Research Assisant - Rumah Program ORHL 2026

Januari 2026 - Present (5 bulan)

Sumatera Selatan, Indonesia

Universitas Kuningan

1 tahun 5 bulan

GIS Laboratory Assistant

Maret 2025 - Mei 2026 (1 tahun 3 bulan)

Kuningan

Provided practical instruction in GIS and remote sensing applications for forestry students. Key responsibilities included:

Teaching georeferencing, attribute table editing, and spatial queries

DEM analysis: slope, contour, and hillshade generation

Map layout design and cartographic styling

Rainfall data interpolation and spatial trend analysis

Land cover classification and NDVI-based vegetation analysis, Land Surface temperature

Image preprocessing: radiometric and geometric correction of satellite imagery

Integrating remote sensing outputs with GIS for spatial decision-making

Multi-criteria analysis and scoring for disaster risk mapping

Lab sessions were conducted using ArcMap 10.8 , Tuesday to Friday (1–5 PM), focusing on real-world forestry and environmental challenges.

Remote Sensing Research Assistant

Juli 2025 - Februari 2026 (8 bulan)

Kuningan, Jawa Barat, Indonesia

Peer Mentor - Spatial Analysis and Modeling for Forestry

Maret 2025 - Juli 2025 (5 bulan)

Voluntarily mentored classmates in advanced spatial modeling topics for forestry. Guided peers in hydrological modeling (flow direction, accumulation, and watershed delineation), habitat suitability modeling with Maxent, raster-based modeling, and spatial network analysis. Provided instruction on Topographic Wetness Index (TWI), topographic modeling, mathematical operations in spatial analysis, and spectral statistics using ArcGIS. Strengthened leadership and communication skills through hands-on guidance and collaborative learning in a lab environment.

Student Recruitment Team Faculty of Forestry and Environment
Januari 2025 - Mei 2025 (5 bulan)

Simulated Candidate in Assesor Evaluation - Biomass Parameter
Measurement

Februari 2025 - Februari 2025 (1 bulan)

Kuningan Regency, West Java, Indonesia

Participated as a simulated candidate in the assessor evaluation process conducted by BNSP, supporting the faculty's initiative to establish a licensed Professional Certification Institution (LSP) in forestry. Gained hands-on experience with national certification standards and field-based biomass assessment procedures.

TikTok

GIS & RS Content Creator

Maret 2025 - April 2026 (1 tahun 2 bulan)

remote

I develop tutorial-based educational content derived from real research workflows, where ongoing or published studies are translated into step-by-step explanations. These tutorials are delivered through short educational videos, visual collages, and live discussions, making complex geospatial and environmental analyses more accessible.

My activities include:

Explaining research methodologies and analytical results related to remote sensing, GIS, data visualization, and data science in forestry and environmental studies

Demonstrating practical workflows using Google Earth Engine, Google Colab, R, QGIS, ArcGIS, and Python (including raster processing with rasterio)

Visualizing spatial data to support interpretation of environmental phenomena such as vegetation dynamics, water bodies, and land-use patterns

Engaging in interactive discussions through live sessions and direct messages, providing guidance on GIS analysis, research design, and data visualization challenges

Sharing and reviewing recommended scientific papers to support evidence-based learning and critical interpretation

The account has grown to 2,114 followers with 51.24K total likes, reflecting consistent engagement from students and early-career researchers interested in applied geospatial analysis and environmental research.

Forum Pohon Langka Indonesia (FPLI)

2 bulan

QGIS Training Mentor -Pembuatan Peta Survei Pohon Langka
Februari 2026 - Februari 2026 (1 bulan)

Endangered species surveyor

Januari 2026 - Februari 2026 (2 bulan)

Kuningan

Forum Pohon Langka Indonesia (FPLI) | January 21 – February 21, 2026
Gunung Tilu, Kuningan, West Java

Program Components:

GPS & GIS Training – Instructing field data collection using GPS devices and spatial analysis with QGIS software for rare tree mapping

Rare Tree Species Field Survey – Supervising field identification, inventory, and conservation status assessment of threatened tree populations

Visual Design & Science Communication – Mentoring students in creating posters, infographics, and cartographic materials for conservation awareness

Community Conservation Program Design – Facilitating development of tree adoption schemes and public engagement strategies for biodiversity conservation

Ecological Trip Guide

Januari 2026 - Januari 2026 (1 bulan)

Gunung Tilu Kuningan West Java

GIS and Remote Sensing Analyst

Januari 2026 - Januari 2026 (1 bulan)

I developed land cover, forest area, and landslide susceptibility maps in the Bukit Barisan region using Google Earth Engine and QGIS to support a village rehabilitation program. This work aims to support environmental planning, risk mitigation, and sustainable land management in mountainous and forested areas.

Behance

Freelance Graphic Designer

April 2021 - Mei 2025 (4 tahun 2 bulan)

Kementerian Lingkungan Hidup dan Kehutanan

Social Forestry Facilitator - BPSKL Sumatera Region

September 2024 - Desember 2024 (4 bulan)

Sumatera Utara, Indonesia

As a Social Forestry Facilitator at BPSKL (Social Forestry and Environmental Partnership Agency), I empower local communities to sustainably manage their forest resources while fostering environmental conservation. My responsibilities include supporting the formation and capacity-building of Social Forestry Business Groups (KUPS), assisting in the development of Social Forestry Management Plans (RKPS) and annual work plans (RKT), and guiding communities in understanding and applying social forestry policies. I also contribute to boundary marking and tenure clarification to secure community-managed forest areas, while building partnerships with government agencies, NGOs, and private sector stakeholders. Through collaboration, knowledge-sharing, and problem-solving, I aim to contribute to a future where forests are sustainably managed, biodiversity is preserved, and communities thrive.

I successfully completed the MSIB Batch 7 JaLan MaPan Program as a Social Forestry Facilitator for the Muara Indah Jaya Mandiri Group in Denai Kuala Village, Pantai Labu Subdistrict, Deli Serdang, North Sumatra. In this program, I developed competencies in:

1. Institutional Analysis: Strengthened group structure and governance through evaluations.
2. Area Analysis: Conducted surveys, mapping, and inventory of non-timber forest product (NTFP) potentials.
3. Policy Analysis: Ensured alignment with social forestry regulations.

4. Business Analysis and Product Innovation: Developed innovative product ideas to enhance productivity.
5. Management Plan Development: Drafted Social Forestry Work Plan (RKPS) and Annual Work Plan (RKT) for resource management.
6. Community Empowerment: Organized training to promote sustainability and long-term impact.

This experience refined my skills in forestry management, policy alignment, and community engagement, deepening my dedication to advancing sustainable development through social forestry initiatives.

Universitas Kuningan

1 bulan

Graphic Designer | Fahutan Expo X Super Music

Januari 2024 - Januari 2024 (1 bulan)

Kuningan, Jawa Barat, Indonesia

FAHUTAN EXPO 2024 was a significant event organized as part of a Project-Based Learning (PJBL) initiative by students enrolled in the Non-Timber Forest Products (HHBK) course. The event aimed to highlight the diverse potential of forests beyond just timber, emphasizing the ecological and economic value of non-timber forest products. It raised public awareness about the vast resources forests offer and promoted sustainable management practices.

Graphic Designer - Faculty of Forestry and Environment, Universitas Kuningan

Januari 2024 - Januari 2024 (1 bulan)

As a Graphic Designer for the Faculty of Forestry and Environment at Universitas Kuningan, I contributed to enhancing the faculty's visual identity and promotional efforts. My responsibilities included:

This role allowed me to combine my passion for design with my academic background, helping the faculty effectively communicate its vision and attract a broader audience.

Pendidikan

Universitas Kuningan

S.Hut, Forestry · (2022 - 2026)

Madrasah Aliyah Negri 1 Ciamis

Mathematics and natural science · (Juli 2019 - Juni 2022)

Pondok Pesantren Darussalam Ciamis

Studi Agama Islam · (Juni 2019 - Juni 2022)

Pondok Pesantren Modern Al Ihya Cigugur Kuningan

Pendidikan Agama Islam · (Juni 2017 - Juni 2019)

Madrasah Tsanawiyah 3 Kuningan

· (2017 - 2019)